Control System Training

MODULE 11 – Position Control

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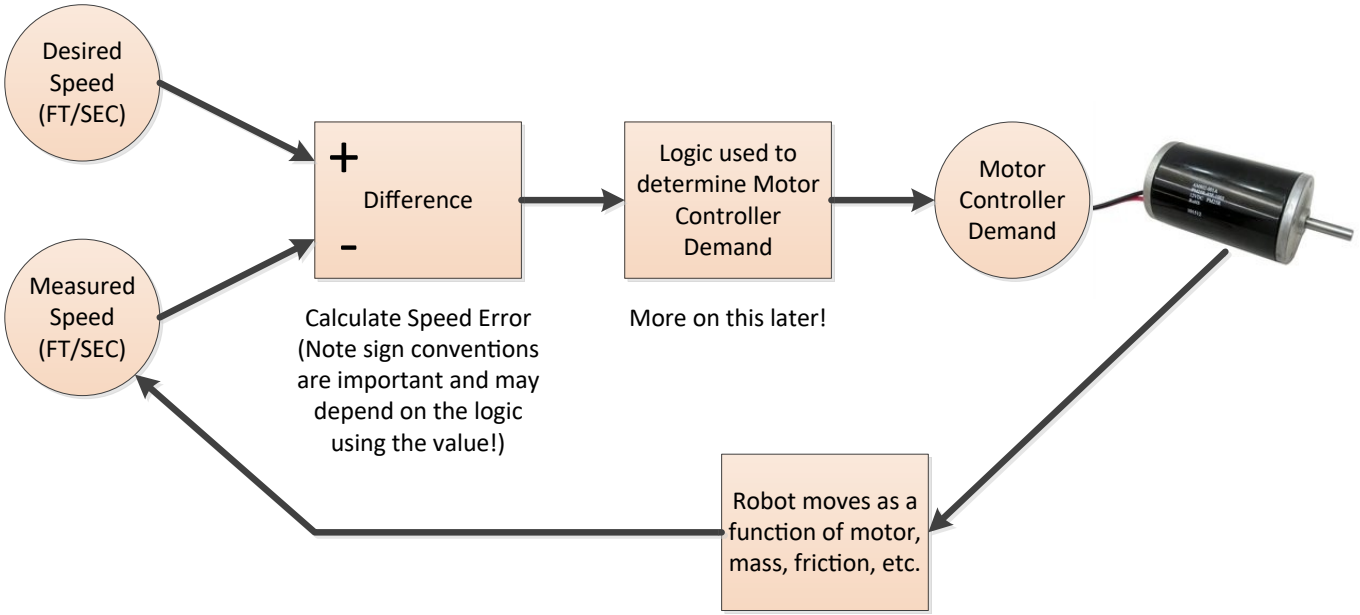
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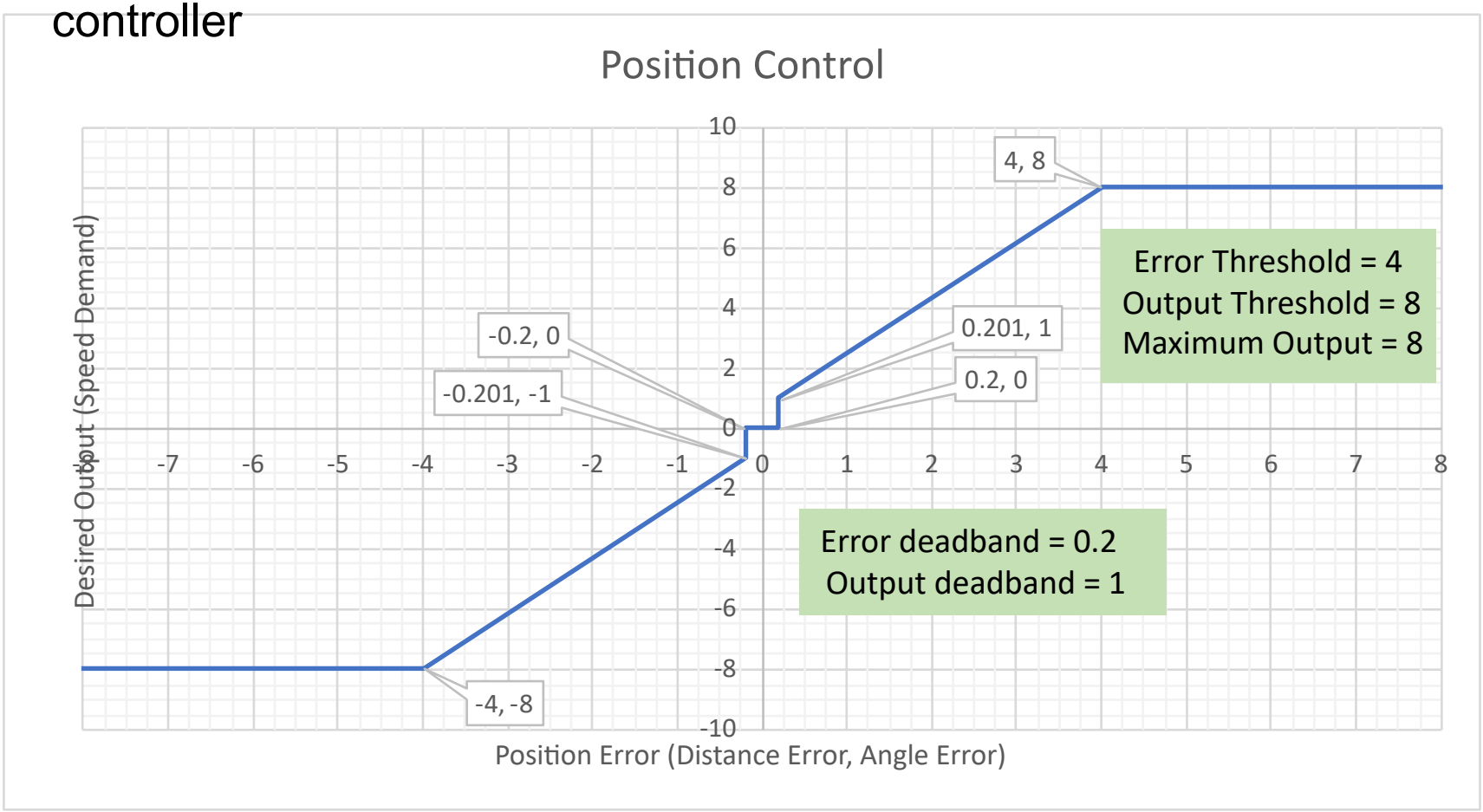
Position Control Definition

- **A form of Analog Closed Loop Control Logic**
  - 

- **Can be used ONLY when CO is always zero when Control Error is zero**
  - Examples: arm position, turret position, swerve drive wheel angle position, spin robot, drive straight a certain distance.
  - Can be used to directly set CO output (arm position, turret postion) or can be used to generate a SP for a cascaded controller to use (drive straight).

Position Control Algorithm

- **X axis –** Control Error (SP – PV)
- **Y axis –** Control output (CO) or Setpoint (SP) to cascaded controller



Position Control Algorithm

- **Inputs**
  - Control error (SP – PV) Units don't matter, but must match
- **Outputs**
  - Control output (CO) Units don't matter.
- **Tuning Parameters (All positive, automatically negated to apply to negative control errors.)**
  - **Error deadband** – Defines accuracy of our position.
  - **Output deadband** – Define as smallest output that will create movement.
  - **Error threshold, Output threshold** – These two values define the slope of the output slow down as the error approaches zero. Larger slopes will get to target faster, but may cause oscillation. Smaller slopes will take longer to get to target.
  - **Maximum output** – Defines the maximum value of the CO. This will clamp the output

Position Control Algorithm Advanced

- **Tuning Parameters (advanced)**
  - **Kd –** Derivative constant. Multipled to the change in control error / change in time (derivative). This can be used to dampen oscillations. A very small value 1.0E-4 does a lot!
  - **Filter derivative –** true/false. When true the change in error is averaged over the last 3 scan periods.
- **Simplified algorithm equation**

```
if ( ABS( error ) <= error_deadband ) then
        out = 0.0
else
        out = error * slope + intercept + Kd * d(error)/dt
endif
CO = clamp( out, max_output )
```

  - Slope and intercept are different for negative and positive errors.
  - Clamping occurs in negative and positive direction.

Position Control – Tuning

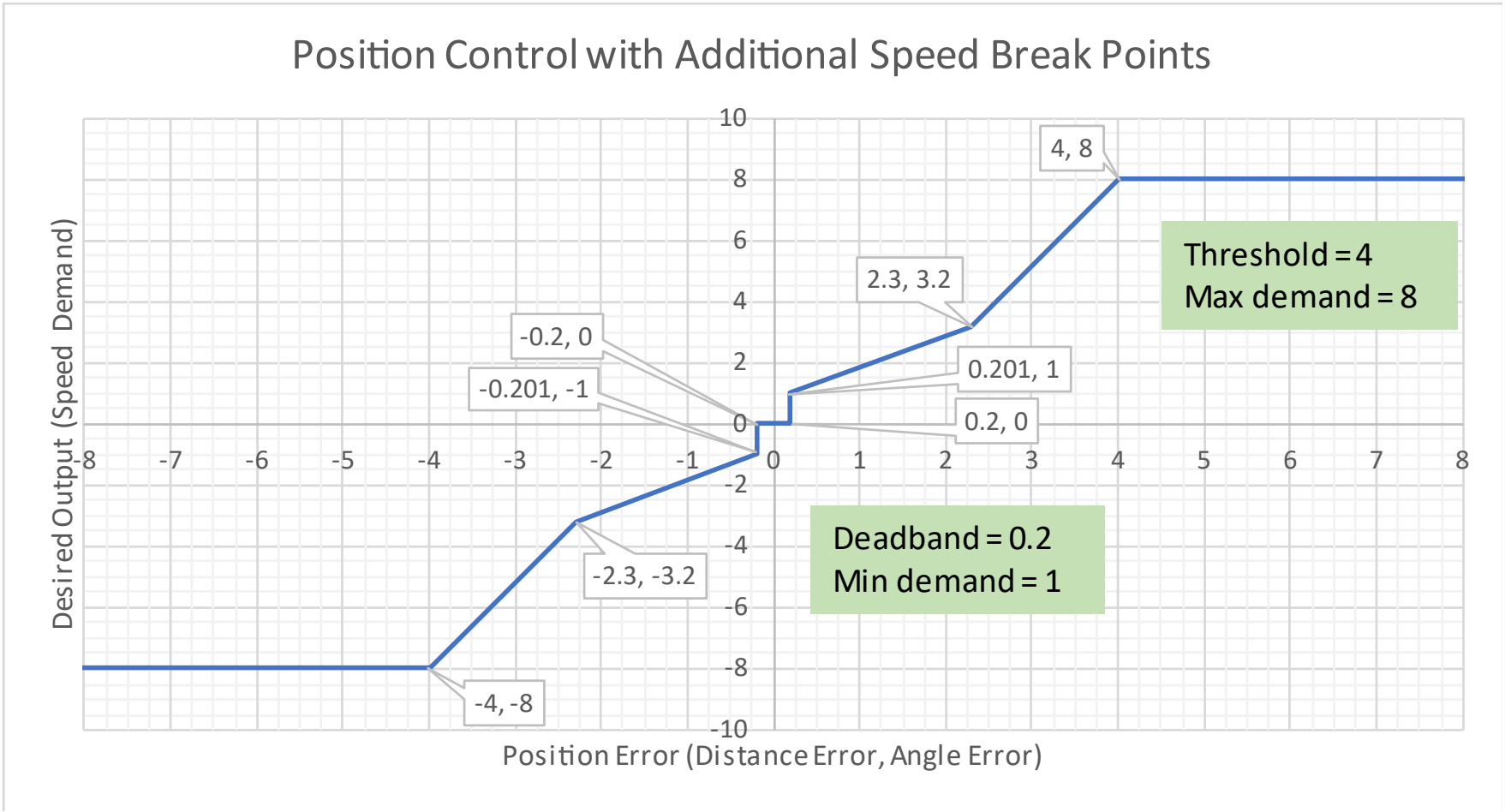
- **Set initial tuning, then test and adjust to minimize oscillation and maximize speed**
- **Initial tuning:**
  - **Error deadband** – Suggest setting to be between 0.5 and 1 times the accuracy needed. Likely won't need to be adjusted.
  - **Output deadband** – Guess at the smallest output that will make the system move.
  - **Error threshold** – Start with this fairly large. As testing progresses make this smaller until oscillation happens.
  - **Output threshold –** Start with this small. As testing progresses make this larger to speed up movement.
  - **Output maximum** – Set equal to Output threshold.
  - **Kd –** Start with this small, or zero. Increase slowly to help dampen system. This can help speed up control too, but with the potential greatly increase oscillation.

Position Control – More Tuning

- **Tuning takes time. This should be built into the schedule.**
- **Keep a record of all the parameters that were tried and what happened each time.**
- **Test different conditions. For example tuning parameters for spinning a robot drive will work differently on carpet than on tile!**
- **Consider that the system may wear over time. Don't set the output deadband too small.**
- **Make it easy to tune. Put the tuning parameters into network table variables that can be set from the dashboard.**

Alternate Position Control Algorithm

- **Possible alternate version with additional error, speed demand break point pairs**



Questions

Questions

Exercise 11.1

- **Write the algebraic expressions for the position control algorithm output when:**
  - Error >= threshold
  - Error <= -threshold
  - Absolute value of error < deadband
  - Error >= deadband and error < threshold
  - Error <= -deadband and error > -threshold
- **The output should be a function of:**
  - Maximum speed
  - Minimum speed
  - Zero
- **Question: What controls the speed from 0 to the maximum speed when the position control is first started?**
- **Question: What analog algorithm could be used to implement this?**

Exercise 11.2

- **Write a generic state machine to spin the robot.**
- **The inputs are:**
  - Maximum speed (Ft/Sec)
  - Amount to spin (Degrees) (+ = clockwise, - = counterclockwise)
  - Timeout time (seconds)
  - Consecutive
- **Outputs (while in progress)**
  - Wheel speed demand (FT/SEC)
- **Outputs (when completed)**
  - On target digital (If not on target, then timeout occurred.)
- **Hint: This state machine only has 3 states.**